Fiscal Year:	FY 2005	**Task Last Updated:**	FY 04/19/2005
PI Name:	Fuller, Charles A. Ph.D.		
Project Title:	Vestibular Modulation of the Circadian Timing System in Altered Gravity Environments		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
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Contact Monitor:		**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Grant number changed from NNA04CC82A to NNA04CC82G per A. Chu/ARC (6/07)		
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Organisms have evolved in Earth's static gravitational environment. Thus, it is perhaps not surprising that exposure to non-Earth gravitational fields can influence an organism's physiology and behavior. One of the affected physiological systems is the Circadian Timing System (CTS). The CTS is responsible for coordinating an organism's physiology and behavior, both internally and with the external 24-hour day. Proper CTS function is essential for maintaining normal physiology and behavior. Recent work in our laboratory has revealed that the vestibular system plays a role in mediating the CTS responses to changes in gravity. This program will elucidate the physiological and molecular mechanisms by which the vestibular system influences the CTS in an altered gravity environment. Our overall objective is to determine if the changes in circadian function induced by exposure to a hyperdynamic environment (i.e. 2G) are mediated by input from the vestibular sensory receptors located in the inner ear, specifically the macular receptors. In order to accomplish

Task Description:	this objective, this task will elucidate the physiological and molecular influence of the vestibular sensors on the regulation of the endogenous circadian period in +G environments. In addition, we will determine the role of vestibular inputs on the regulation of the physiological and molecular responses of the circadian pacemaker to photic stimuli in +G environments. The circadian timing system (CTS) is essential for the generation and internal temporal coordination of biochemical, physiological and behavioral rhythms, as well as for synchronizing these physiological oscillations with the external 24-hour environment. When CTS function is disrupted, an organism's ability to respond appropriately to a physiological stressor is inhibited. Patients with CTS dysfunction often suffer from sleep-wake, affective and neuroendocrine disorders. In addition, CTS dysfunction can severely compromise an individual's long-term health. The vestibular system is responsible for perception, orientation and equilibrium within the three dimensional environment. Over the last 10 years a wider functional role has been demonstrated for the vestibular system. The vestibular system can modulate cardiovascular and respiratory function and has a role in anxiety disorders and spatial navigation. We have recently described a role for the vestibular system in the regulation of physiological homeostasis and circadian function. Our objectives are aimed at understanding how the vestibular system mediates CTS responses to changes in gravity. Vestibular mediated alterations in the period of the pacemaker would affect the ability of the CTS to synchronize with the external environment. Any influence of the vestibular system on photic responsiveness of the CTS would have important implications for both astronauts in altered gravity environments as well as individuals with vestibular and/or circadian disorders.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although considerable knowledge concerning the physiology of the vestibular system and the Circadian Timing System (CTS) has been gained over the past 10 years, little is known about the interaction of the two systems in altered gravity. The proposed studies will reveal fundamental aspects of the physiology of both systems. Circadian rhythms are fundamental regulatory processes of the nervous system and, as such, the putative ability of the vestibular system to modulate the CTS is of great potential significance with clinical and basic science applications. By understanding the physiological and molecular mechanisms by which the vestibular and circadian systems interact, therapeutic and diagnostic strategies may be developed to alleviate vestibular circadian and sleep-wake disorders with vestibular and/or circadian bases. Because of the ubiquitous endogenous control the CTS has on physiology and behavior, this research has a high degree of relevance to a number of disorders, including ones with no prior linkage to vestibular function.
Task Progress:	We have initiated experiments aimed at carrying out our first task objective, that of determining the physiological and molecular influence of the vestibular sensors on the regulation of the endogenous circadian period in +G environments. We continue to gather, process and analyze data.
Bibliography Type:	Description: (Last Updated: 12/08/2025)
Articles in Peer-reviewed Journals	Fuller PM, Jones TA, Jones SM, Fuller CA. "Evidence for macular gravity receptor modulation of hypothalamic, limbic and autonomic nuclei." Neuroscience. 2004;129(2):461-71. PMID: 15501603 , Sep-2004